Dynamic Traffic-Aware Network Routing and Self-Healing System using Dijkstra's Algorithm

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***Abstract*—In this paper, a simulation of a dynamic communication network utilizing Dijkstra's Algorithm for routing will be created utilizing the MATLAB platform. This proposed framework will differ from the conventional static routing techniques, as it will dynamically respond to network congestion and failures due to the ability to change the weights of the network during runtime. It will also be able to perform self-healing due to the ability to dynamically change the routing paths due to node and link failures. Other factors will be implemented to make the simulation more realistic and to make it easier to evaluate the performance of the network.**

***Index Terms*—Dijkstra Algorithm, Dynamic Routing, Self-Healing Networks, MATLAB, Congestion Control.**

I. Introduction

The current communication system has developed rapidly with the advent of increasing internet usage, cloud computing, and IoT-based applications. The current static routing technique does not consider the changes in real-time conditions, leading to inefficient communication.

In order to overcome the issues faced by the current static routing technique, there is a need to develop a dynamic routing technique. In this paper, a traffic-aware dynamic routing technique based on Dijkstra's Algorithm is implemented. The routing technique is able to update the routes based on the conditions, ensuring the best routes. In addition, self-healing is also incorporated in the routing technique, ensuring reliability.

A. Motivation

This is because the increase in IoT devices, real-time applications, and cloud services has made network traffic dynamic and unpredictable. Static routing is unable to efficiently handle such dynamic changes. Hence, it is important to use adaptive routing techniques to ensure that bandwidth is utilized in an efficient way and that communication is not interrupted.

B. Contribution

- Dynamic network generation.
- Interactive congestion and failure simulation.
- Self-healing routing mechanism.
- Bottleneck Bandwidth and convergence time analysis.

II. Objective

- Implement Dijkstra Algorithm based shortest path routing.
- Simulate congestion, link failures and node failures.
- Initiate dynamic and adaptive routing.
- Analyse performance using key metrics such as cost and convergence time.

III. System Model

The communication network is modeled as a graph:

$$G = (V, E)$$

where V represents nodes (routers) and E represents links. The cost of a path is defined as:

$$C(P) = \sum w(u, v)$$

where $w(u, v)$ represents the weight of the link between nodes u and v. These weights dynamically change based on congestion and failures.

IV. Methodology

A. Network Generation

First, a connected base topology is created to ensure that all nodes are connected. Additional random connections are made to simulate a realistic mesh network. This guarantees redundancy and multiple alternative routes for fault tolerance.

B. Routing Algorithm

A customized implementation of Dijkstra's Algorithm is used to compute shortest paths.

Detailed Steps:

1) Initialize all node distances to infinity from the source node.
2) Assign distance zero to the source node.
3) Insert nodes into a priority queue.
4) Select node with minimum distance.
5) Perform edge relaxation:

$$d(v) = d(u) + w(u, v)$$

6) Update distances and predecessor nodes.

7) Repeat until all nodes are processed.
8) Reconstruct shortest path.
9) Recompute paths dynamically when network conditions change.

C. Dynamic Features

- **Congestion:** Link cost is increased to simulate real traffic load.
- **Link Failure:** Link weight is set to ∞ to simulate real link failure.
- **Node Failure:** Node and its connections are removed.
- **Dynamic Routing:** Paths are recalculated in real time.

D. Interactive Feature Description

The system supports user interaction:

- **Option 1: Traffic Congestion** Simulates heavy traffic by increasing link cost.
- **Option 2: Link Failure** Removes a link by assigning infinite cost.
- **Option 3: Change Source/Destination** Allows testing different routing scenarios.
- **Option 4: Node Failure** Simulates router failure and triggers rerouting.
- **Option 0: Exit** Terminates simulation.

V. Operation

A. Dynamic Behaviour

The system maintains the link weights based on the conditions in the network, such as congestion. Whenever the link weight changes, the algorithm is executed again to find the new optimal path.

B. Failure Recovery

Failure recovery is achieved by assigning infinite weight or deleting nodes from the network. The system automatically recognizes the failure of nodes or links and finds the alternative route, thus maintaining uninterrupted communi- cation.

C. Topology Module

This module generates a connected topology with extra random links, simulating real-world conditions.

D. Cost Matrix Module

This module maintains the link costs dynamically based on network conditions.

E. Routing Engine

This module applies Dijkstra's algorithm, which finds the shortest paths.

F. Visualization Module

- Green: Optimal path
- Red: Failed or congested links
- Blue: Normal links

VI. Results

Since the number of nodes is variable, results are demonstrated using a 10-node network example.

A. Initial Network Topology (10 Nodes)

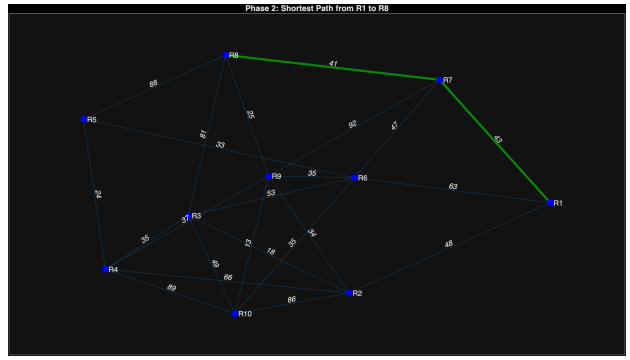

Fig. 1. Randomly Generated Network Topology (10 Nodes)

The generated topology ensures connectivity and provides multiple alternate paths, improving reliability.

B. Shortest Path (R1 to R8)

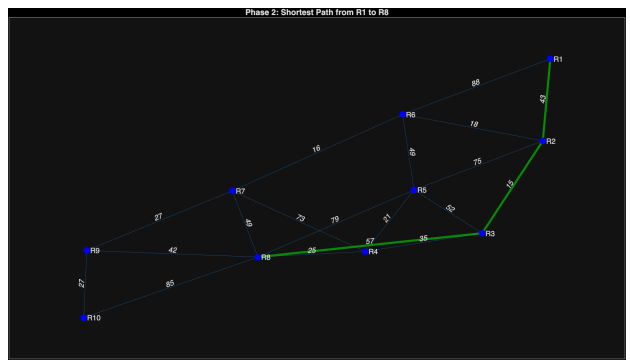

Fig. 2. Shortest Path between R1 and R8

The algorithm computes the minimum cost path between source and destination nodes and highlights it.

C. Congestion Scenario (R3-R5)

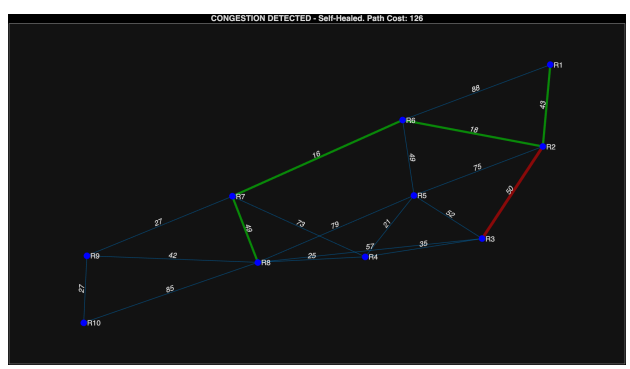

Fig. 3. Routing Adaptation under Congestion

When congestion is introduced, link cost increases and the algorithm avoids the congested path which demonstrating adaptive routing behavior.

D. Link Failure (R2-R6)

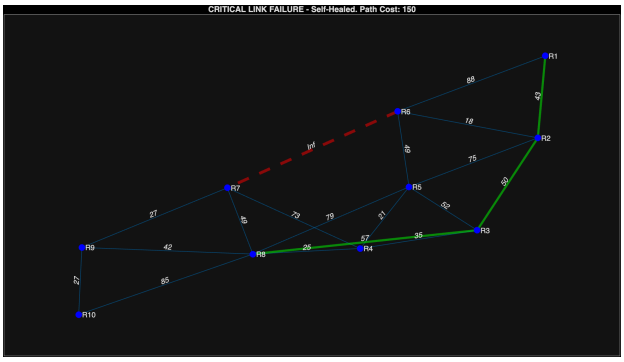

Fig. 4. Routing after Link Failure

After link removal, the system recalculates the path and selects an alternate route.

E. Node Failure (R4)

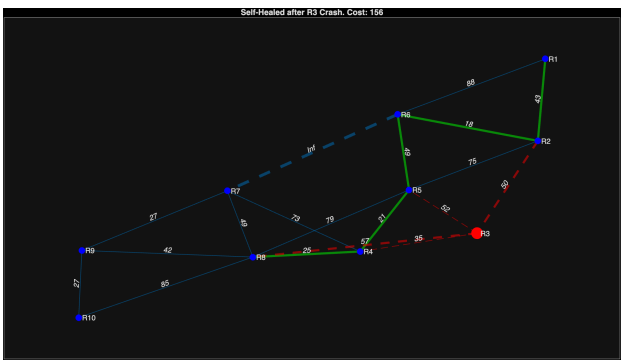

Fig. 5. Self-Healing after Node Failure

This shows that the network is self-healing.

F. Performance Metrics

- The path cost is increased in the case of congestion.
- Bottleneck bandwidth is used to analyze the performance.
- Convergence time is still low.

VII. Performance Discussion

The dynamic routing algorithm enhances the network's strength and efficiency. The network can adapt to failures and congestion quickly. The algorithm ensures that data is transmitted efficiently.

VIII. Limitations

- The model is not tested using real-world traffic.
- The implementation of the protocol is missing.

IX. Future Scope

- The model can be enhanced to use machine learning techniques.
- Real-time traffic can be incorporated into the model.
- The model can be enhanced to have a graphical user interface.

X. Conclusion

This project successfully demonstrates how a dynamic and self-healing routing system works using Dijkstra's Algorithm. The system is efficient and adapts to network changes such as congestion and network failures.

References

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[3] IEEE Communications Society, 2023.